

$$\begin{aligned}
&\bullet \quad \theta \qquad \qquad \qquad \pi_\theta(y \mid x) \\
&\bullet \quad \pi \\
&\bullet \quad r_i(\theta) = \frac{\pi_\theta(y_i \mid x_i)}{\pi(y_i \mid x_i)} \qquad \qquad \qquad (x_i, y_i) \\
&\bullet \quad r_j^{(k)}(\theta) = \frac{\pi_\theta(y_{j,k} \mid x_j)}{\pi(y_{j,k} \mid x_j)} \qquad \qquad \qquad k \qquad \qquad \qquad x_j
\end{aligned}$$

$$\begin{aligned}
&\bullet \quad \hat{A}_i \qquad \qquad \qquad \pi(y_{j,k} \mid x_j) \\
&\bullet \quad w_{j,k} \geq 0 \qquad \qquad \qquad y_{j,k} \qquad \qquad \qquad \sum_k w_{j,k} = 1 \qquad \qquad \qquad j \\
&\bullet \quad \text{Clip interval parameter} \quad \epsilon > 0 \qquad \qquad \text{clip}_\epsilon(r) = \text{clip}(r, 1 - \epsilon, 1 + \epsilon) \\
&\bullet \quad L_{\text{CPO}} \\
&\bullet \quad D_{\text{KL}}(\pi \parallel \pi_\theta)
\end{aligned}$$

$$\begin{aligned}
J_{\text{HACPO}}(\theta) = & \mathbb{E}_{(x_i, y_i) \sim \mathcal{D}} \left[\underbrace{\alpha_i \cdot \min(r_i(\theta) \hat{A}_i, \text{clip}_\epsilon(r_i(\theta)) \hat{A}_i)}_{\text{HACPO}} \right. \\
& \left. + (1 - \alpha_i) \cdot \underbrace{\sum_{k=1}^K w_{i,k} \min(r_i^{(k)}(\theta) \hat{A}_i^{(k)}, \text{clip}_\epsilon(r_i^{(k)}(\theta)) \hat{A}_i^{(k)})}_{\text{HACPO}} \right]
\end{aligned}$$

$$\begin{aligned}
&+ \gamma \mathbb{E}_{x_i \sim \mathcal{D}, \{y_{i,k}\} \sim A} [L_{\text{CPO}}(\theta; x_i, \{y_{i,k}\})] \\
&- \beta_{\text{KL}} D_{\text{KL}}(\pi \parallel \pi_\theta).
\end{aligned}$$

$$\begin{aligned}
&\beta_{\text{KL}} D_{\text{KL}}(\pi \parallel \pi_\theta) \qquad \qquad \qquad J_{\text{HACPO}} \qquad \qquad \qquad J_{\text{HACPO}} \\
&D_{\text{KL}}(\pi \parallel \pi_\theta) \qquad \qquad \qquad \pi_\theta
\end{aligned}$$

$$L_{\text{CPO}}$$

